SIDD-Medium Denoising Results

Tanika (@Tanika-git) | U-Net and NAFNet | 29 October 2026

Experiment summary: 10 baseline notebook sections and genuine two-step CPU smoke-test results. The selected Python excerpts are explanatory, not runnable cells.

1. Mount Drive and edit paths

The complete SIDD-Medium sRGB collection was verified: **160 scene-instance folders, 320 noisy/GT image pairs, 640 PNGs**. The private Drive folder, credentials and exact paths are intentionally not included.

2. Get code and inspect runtime

Both models use the credited public baseline repository. Experiments ran on CPU. The model construction interface was:

```
from models import build_model
model = build_model(config["model"])
```

3. Stage the dataset

The original notebook supports full staging to local Colab disk. For this short demonstration, full staging was **not performed**. Three scene-separated image pairs were prepared as small crops to avoid another large data transfer.

4. Verify pairs and fixed splits

Three distinct physical scenes supplied one pair each, assigned using a fixed random seed to **1 train / 1 validation / 1 test** image. This tiny set is for debugging only, not for statistically meaningful conclusions.



5. Preserve experiment configurations

Setting	U-Net and NAFNet
Training device	CPU
Training steps	2 each
Batch size	1
Training patch	64 x 64 pixels
Sample inference	64 x 64 center crop

```
tiny['batch_size'] = 1
tiny['patch_size'] = 64
tiny['max_steps'] = 2
```

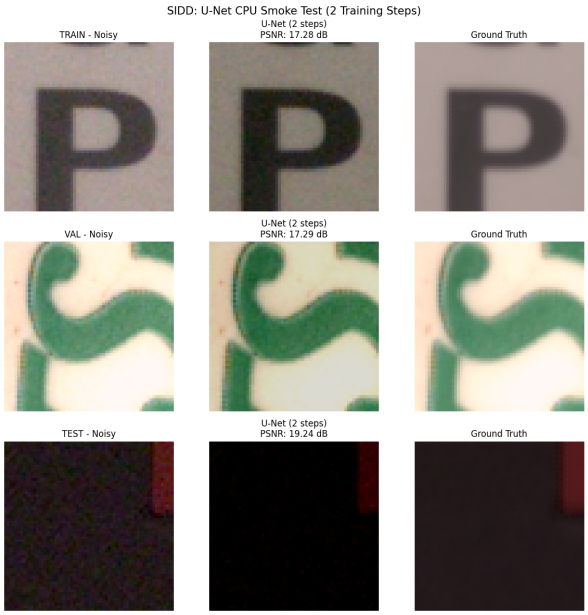
6. Short sanity run for both models

Both models executed the two-step smoke test on real SIDD pairs, saved checkpoint files, and generated image predictions. **Passing the smoke test does not establish restoration quality.**

7. Train U-Net - preliminary execution

Two optimizer steps completed. Checkpoint restoration was verified. The training script reported a single-pair validation diagnostic: **PSNR 17.287 dB, SSIM 0.92745**.

```
state = torch.load(checkpoint, map_location='cpu', weights_only=True)
model.load_state_dict(state['model'], strict=True)
model.eval()
```



8. Train NAFNet - preliminary execution

Two optimizer steps completed. Checkpoint saving and inference were verified. The training script reported **PSNR 11.306 dB, SSIM 0.41277** on the single-pair validation diagnostic. Current predictions show visible color/grid artifacts.

9. Evaluate and save results

The original full tiled-evaluation cell was **not run**. These are the actual PSNR measurements on the **same 64 x 64 crop** for each method, not dataset-wide averages.

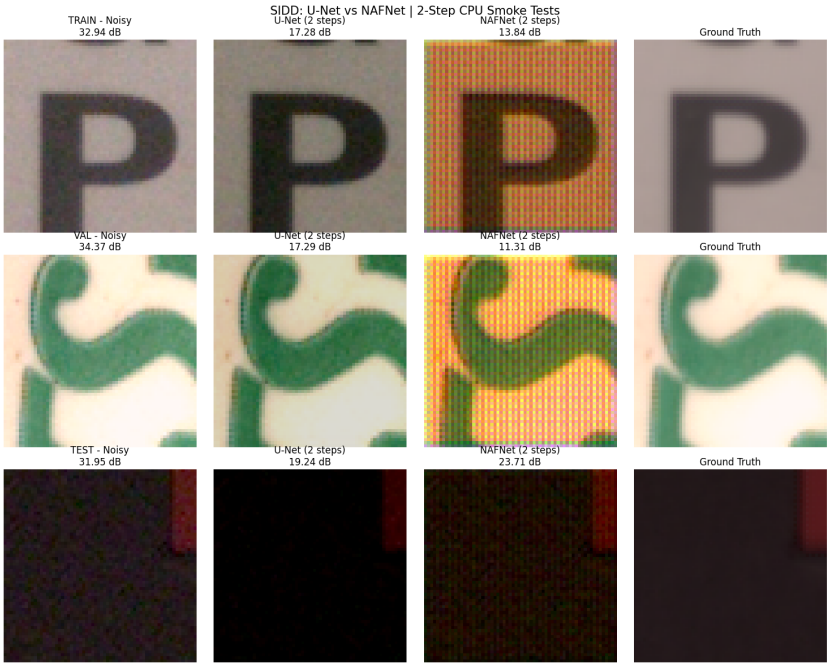
Crop	Noisy	U-Net (2 steps)	NAFNet (2 steps)
Train	32.94	17.28	13.84
Validation	34.37	17.29	11.31
Test	31.95	19.24	23.71

Interpretation: all two-step predictions are worse than their noisy inputs on these particular crops. The test example has already been inspected, so final research needs an untouched holdout set.

```
with torch.inference_mode():
    predicted = model(noisy_tensor)
```

10. Preview and share the saved results

The same image crops are compared as noisy input, U-Net output, NAFNet output and matching ground truth.



Rows: training, validation and exploratory test example. Columns: noisy, U-Net, NAFNet, ground truth. All models trained for **only two CPU steps**; these are pipeline diagnostics, not fully trained denoisers.

Next milestone and attribution

Train both models with a consistent protocol and durable checkpoints; verify PSNR/SSIM on proper held-out data; then study a literature-checked gating modification.

Model code: nitya-goyal/SOTA-Image-Denoising (credited baseline). Dataset: SIDD-Medium sRGB, Abdelhamed et al., CVPR 2018. Original runnable code and private experiment storage are **not embedded**.

Results compiled and smoke tests executed by Tanika (@Tanika-git).